Process synchronization

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(partly based on slides of Le Thanh Van)

Outline

1. Basic concepts
2. Critical-section (CS) problem
3. Synchronization hardware
4. Software tools for synchronization
   - Mutex lock
   - Semaphores
5. Synchronization problems

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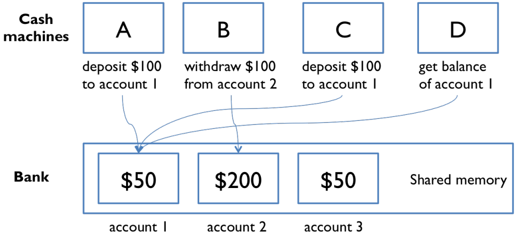
Why do we study synchronization ? (1)

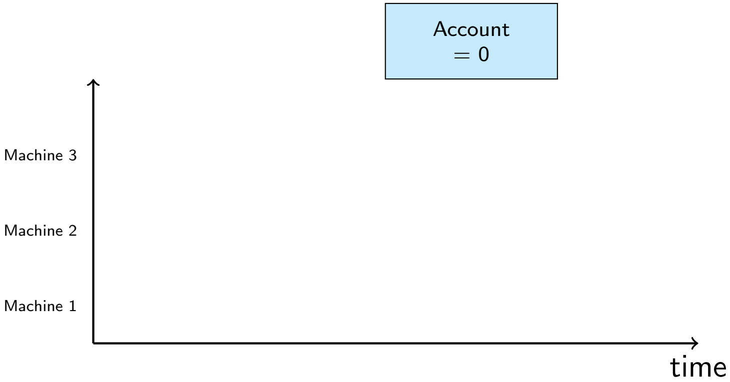
- Different CPU schedulers produce different timings for scheduled processes
- The correctness of a concurrent program should not depend on accidents of timing



(Source: mit.edu)

Cooperating (concurrent) processes which possibly share a logical address can create the inconsistency in shared data

Why do we study synchronization ? (2)



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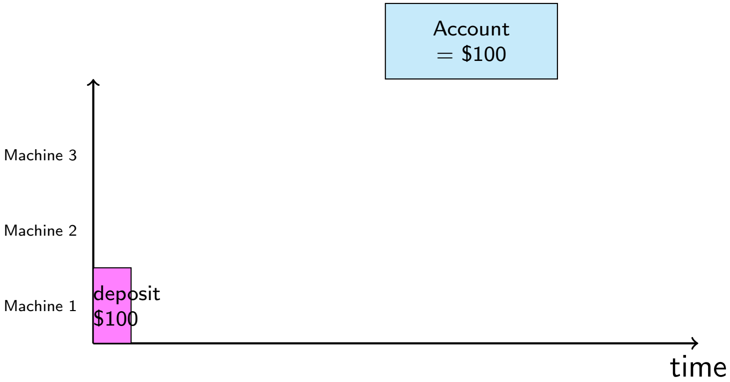

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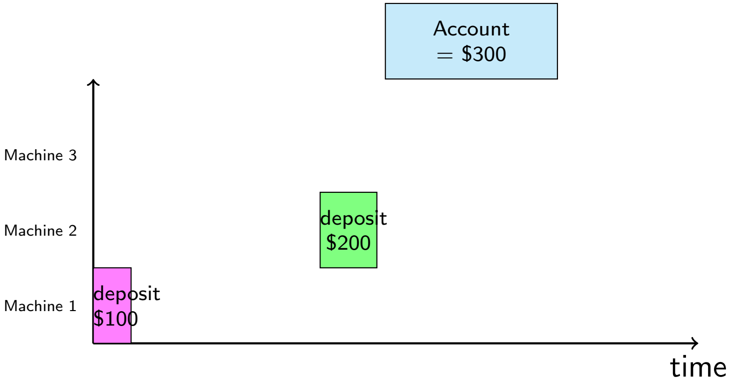

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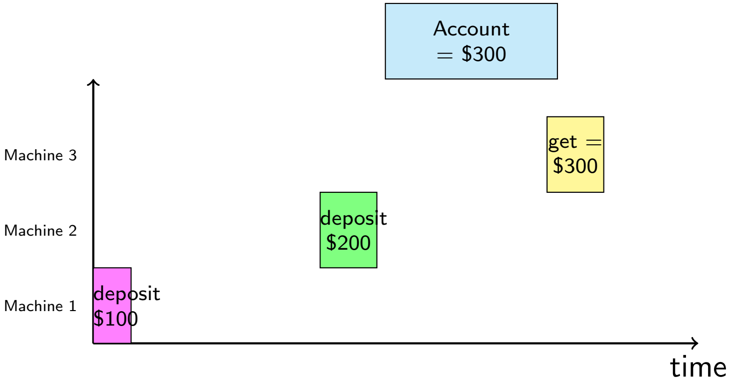

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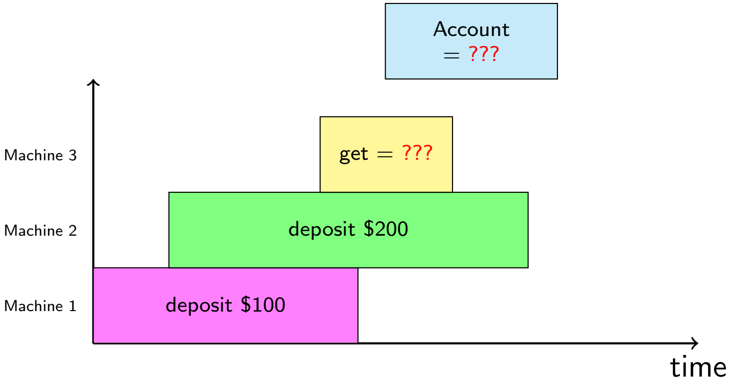

Inconsistency in shared-memory communication (1)

Use a counter to remedy (chữa bệnh) for buffer size of `BUFFER_SIZE - 1`

Producer

```
item next_produced;
while (true){
  while ( counter == BUFFER_SIZE )
    ; /* do nothing */
  buffer[ in ] = next_produced;
  in = ( in + 1 ) % BUFFER_SIZE;
  counter ++;
}
```

Consumer

```
item next_consumed;
while (true){
  while ( counter == 0 )
    ; /* do nothing */
  next_consumed = buffer[ out ];
  out = ( out + 1 ) % BUFFER_SIZE;
  counter --;
}
```

- The codes seem correct, but not
- Suppose currently `counter = 5`, then 2 statements `counter++` and `counter--` executed concurrently
- After those executed, value of `counter` could be 4, 5, or 6 (inconsistently).
  (the expected is 5)

Inconsistency in shared-memory communication (2)

`counter++`

$register_1$ = `counter`
$register_1 = register_1 + 1$
`counter` = $register_1$

`counter--`

$register_2$ = `counter`
$register_2 = register_2 - 1$
`counter` = $register_2$

Concurrent execution of 2 processes is interleaved in a low-level sequential execution in any arbitrary order, perhaps as follows.

T_0:	producer	execute	$register_1$ = `counter`	($register_1 = 5$)
T_1:	producer	execute	$register_1 = register_1 + 1$	($register_1 = 6$)
T_2:	consumer	execute	$register_2$ = `counter`	($register_2 = 5$)
T_3:	consumer	execute	$register_2 = register_2 - 1$	($register_2 = 4$)
T_4:	producer	execute	`counter` = $register_1$	(`counter = 6`)
T_5:	consumer	execute	`counter` = $register_2$	(`counter = 4`)

Inconsistency in shared-memory communication (2)

counter++

$register_1$ = `counter`
$register_1$ = $register_1 + 1$
`counter` = $register_1$

counter--

$register_2$ = `counter`
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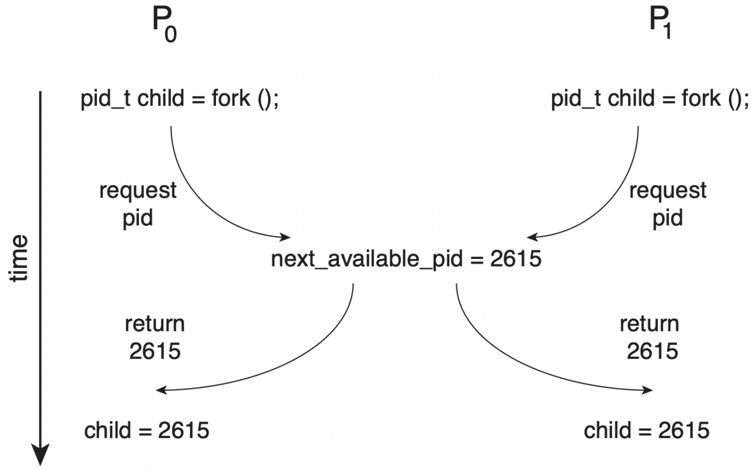
Concurrent execution of 2 processes is interleaved in a low-level sequential execution in any arbitrary order, perhaps as follows.

T_0: producer execute $register_1 =$ `counter` ($register_1 = 5$)

Race condition

Outcome of executing of multiple cooperating concurrent processes depends on particular order of their access events

Race condition example: fork()



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Critical section

Multiple processes should have the following structure to avoid race condition

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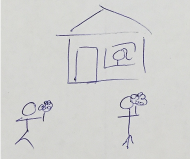
do {
    entry section
        critical section
    exit section
        remainder section
} while (true);
```



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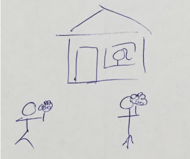

Critical section in kernel-mode process

- Kernel-mode processes share some kernel data structures
  list of opened files
- 2 approaches to handle critical sections in OS
  - Nonpreemptive kernel: kernel-mode process cannot be preempted
    $\rightarrow$ No race conditions
  - Preemptive kernel: kernel-mode process can be preempted
    $\rightarrow$ more responsive, but prone to critical section

Conditions to critical section

- Mutual exclusion: at most one process allowed in critical section
- Progress: if no process in critical section and some (one or more) wish to enter critical section, the selection of which (of these processes) will enter cannot be postponed indefinitely
- Bounded waiting: a limit on number of times which other processes allowed to enter critical section after a process makes a request and before that request granted.

Algorithm 1 for CS

(assumed for case of 2 processes)

- Processes do not know which is in CS
- $\rightarrow$ use a shared variable "`turn`" to know which is in CS. (Initially `turn=0`)

  `turn = i` $\Rightarrow P_i$ is in its CS

Process P_i

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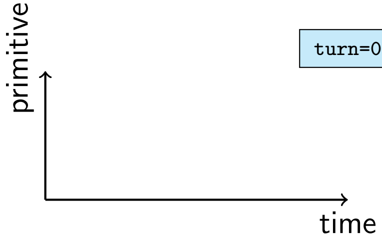

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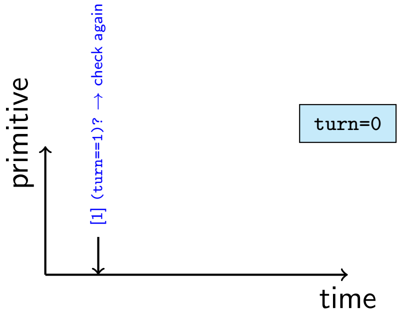

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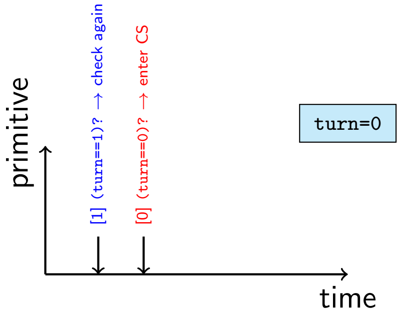

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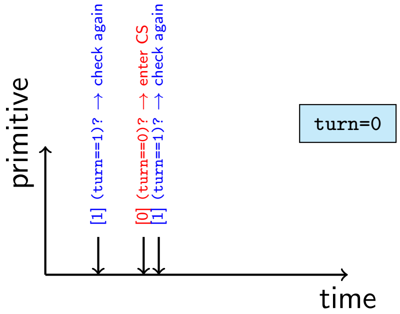

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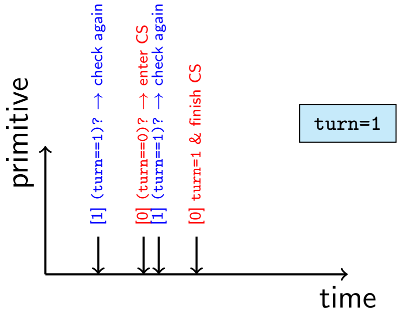

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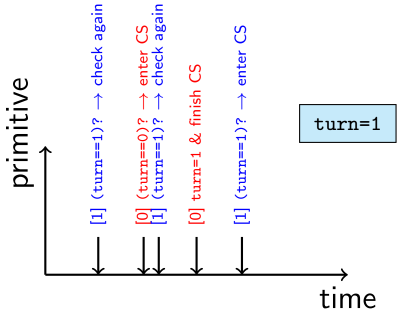

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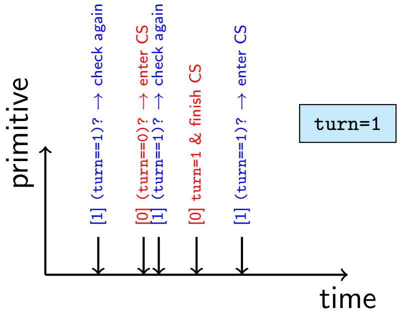

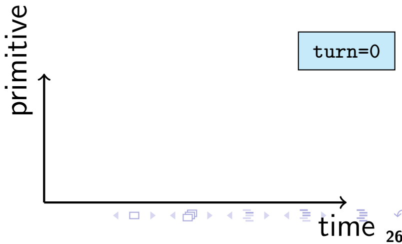

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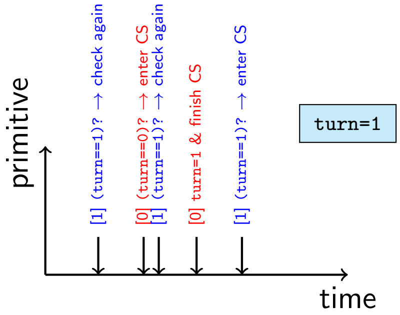

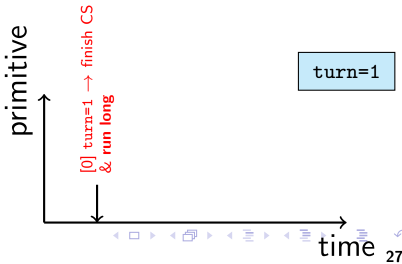

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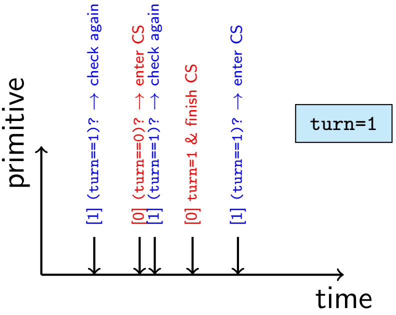

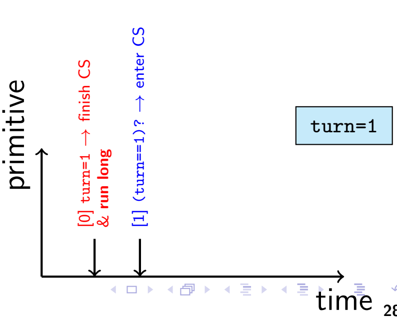

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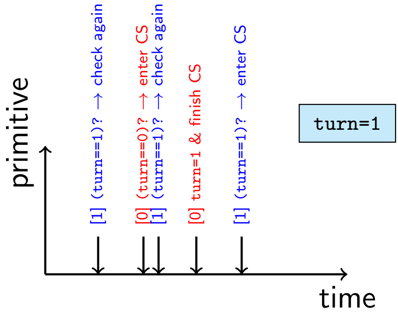

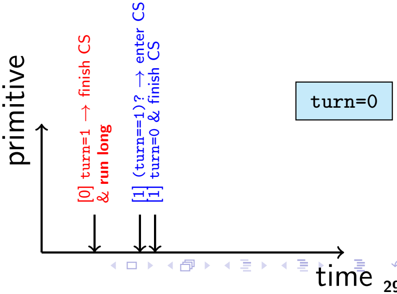

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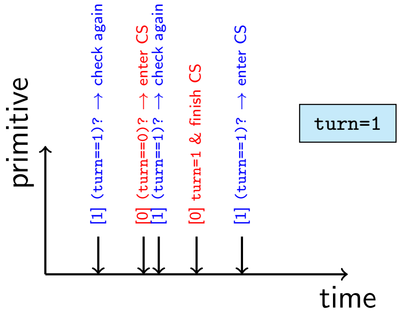

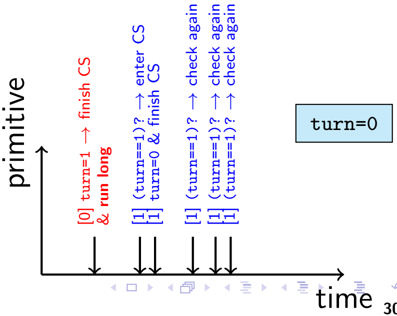

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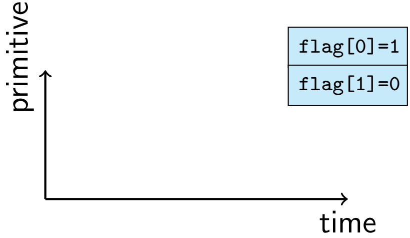

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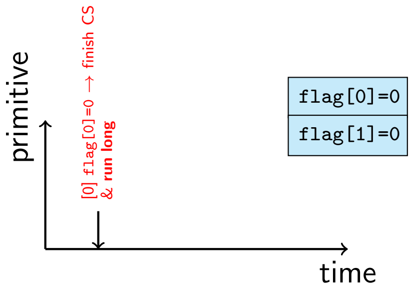

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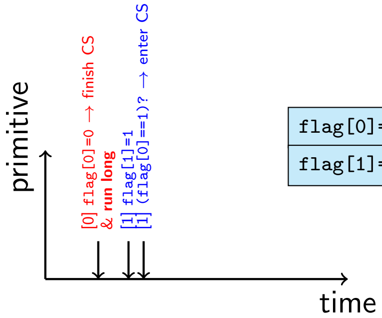

flag[0]=0
flag[1]=1

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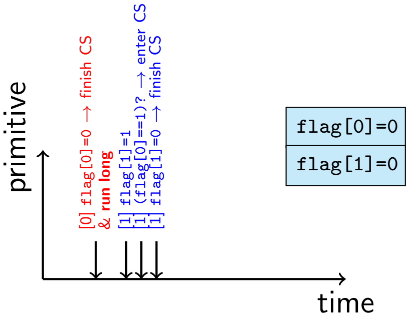

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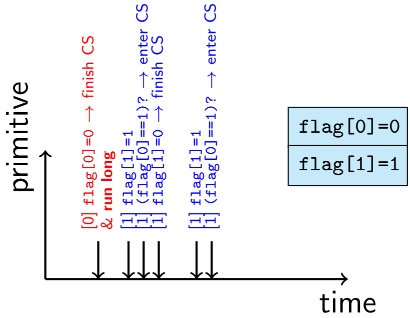

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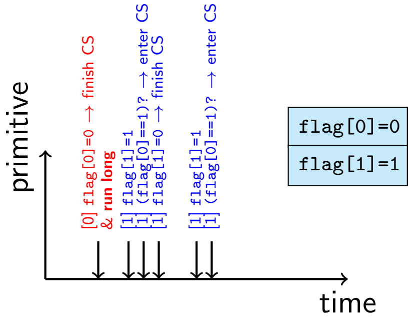

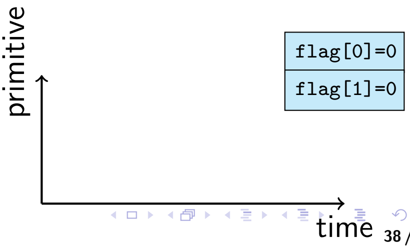

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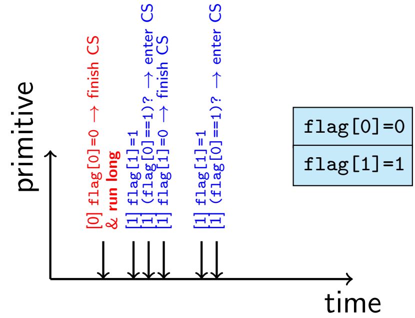

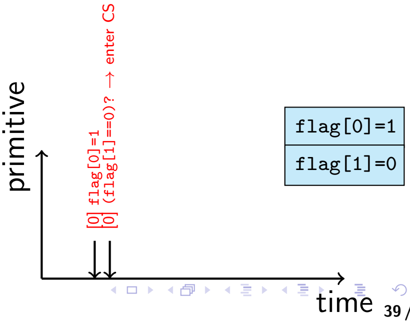

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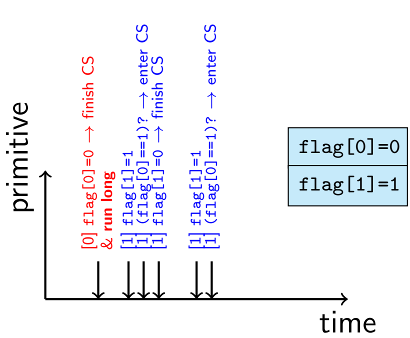
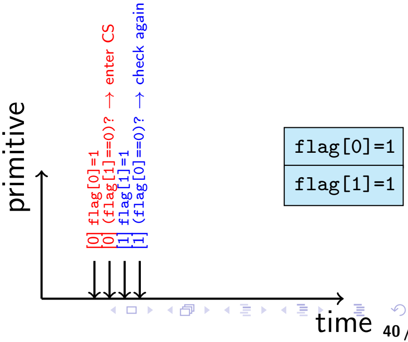
Process P_i

```
do {
  flag[i] = true;
  while ( flag[j] ) ;
     /* critical section code is here  */
  flag[i] = false;
     /* remainder section code is here */
} while(true);
```

mutual exclusion, but not bounded waiting, and not progress





Algorithm 2 for CS

- `turn` cannot keep the state of readiness into CS
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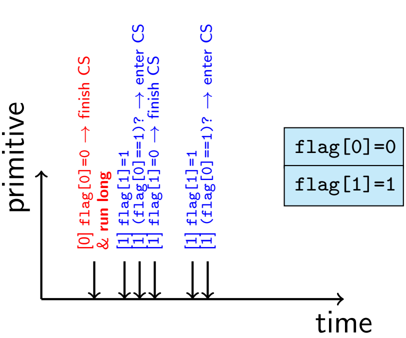

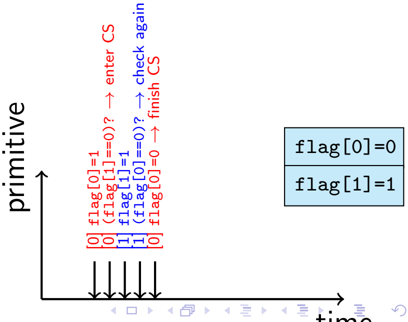

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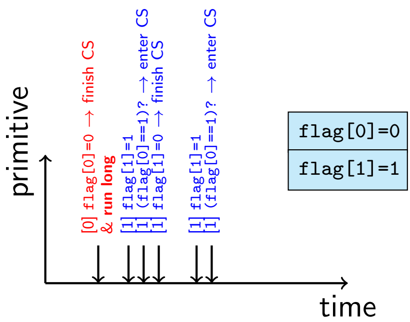

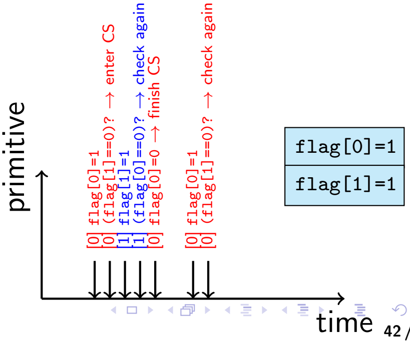

Algorithm 2 for CS

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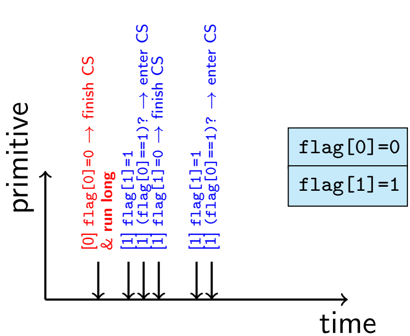
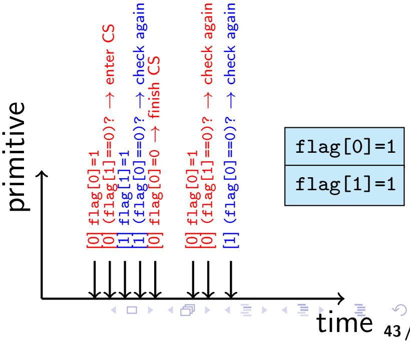
mutual exclusion, but not bounded waiting, and not progress





- Use both kinds of variables of Algorithm 1 & 2

Process P_i

```
do {
  flag[i] = true;
  turn = j;
  while ( flag[j] && turn == j ) ;
     /* critical section code is here  */
  flag[i] = false;
     /* remainder section code is here */
} while(true);
```

Bakery algorithm (1)

Peterson's algorithm is only for 2 processes. Bakery algorithm is for n processes

- Before entering its critical section, process receives a ticket number. Holder of the smallest number enters its CS
- If processes P_i and P_j receive the same number, if $i < j$, then P_i is served first; else P_j is served first
- The numbering scheme always generates numbers in increasing order of enumeration; i.e., 1,2,3,3,3,3,4,5,...

Bakery algorithm (2)

- A process is assigned a pair (pid,ticket #) which is ordered lexicographically
- Shared data
  - `boolean choosing[n]`
  - `number[n]`: ticket #

Process P_i

```
do {
  choosing[i] = true;
  number[i] = max(number[0],number[1],...,number[n-1]) + 1;
  choosing[i] = false;
  for( j=0; j<n; j++ ){
    while( choosing[j] ) ;
    while( number[j] != 0 && (number[j] < number[i] ) ;
  }
     /* critical section code is here  */
  number[i] = 0;
     /* remainder section code is here */
} while(true);
```

Outline

1 Basic concepts

2 Critical-section (CS) problem

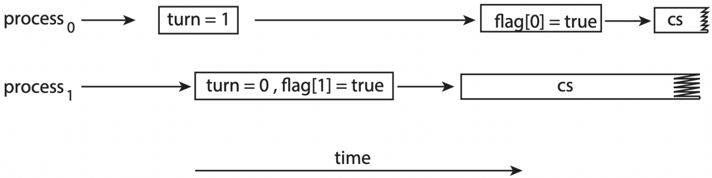
3 Synchronization hardware

4 Software tools for synchronization
- Mutex lock
- Semaphores

5 Synchronization problems

Modern computer architecture

- Multiple cores for multithreaded applications
- Processor or compiler can reorder the execution (if checked no data dependency)



Software-based solutions such as Peterson's are not guaranteed to work on modern computer architectures (mutiple processors/cores)

Hardware synchronization tools

There are 3 forms of hardware instructions to support solving critical-section problem.

- Memory barriers
- Hardware instructions
- Atomic variables

Memory barriers

Memory model

A model that a computer architecture guarantees to application programs on memory.

- **Strongly ordered**: where a memory modification on one processor is immediately visible to all other processors.
- **Weakly ordered**: where modifications to memory on one processor may not be immediately visible to other processors.

Memory barriers

Memory model

A model that a computer architecture guarantees to application programs on memory.

- **Strongly ordered**: where a memory modification on one processor is immediately visible to all other processors.
- **Weakly ordered**: where modifications to memory on one processor may not be immediately visible to other processors.

Memory models vary by processor type. $\Rightarrow$ OS kernel cannot depends on any assumptions regarding the visibility of modifications to memory on a shared-memory multiprocessor.

$\Rightarrow$ Computer architectures provide memory barriers to force any changes in memory to be propagated to all other processors.

Memory barriers - example

Thread 1

```
while (!flag)
  memory_barrier();
print x;
```

Thread 2

```
x = 100;
memory_barrier();
flag = true;
```

`flag` > `x`

`x` > `flag`

The memory barrier will ensure thread 1 will output 100.

We could place a memory barrier into Peterson's algorithm between `flag[i]` and `turn = j`.

Use an uninterruptable (atomic) sequence of instructions which modifies a shared variable `lock` (initially `lock=false`).

test_and_set()

```
boolean test_and_set( boolean *target )
{
  boolean rv = *target;
  *target = true;
  return rv;
}
```

Mutual-exclusion by test_and_set()

```
do {
  while( test_and_set( &lock ) ) ;
     /* critical section code is here  */
  lock = false;
     /* remainder section code is here */
} while (true};
```

Swaping the contents of two words (compare_and_swap()) on a shared variable (lock) (initially lock=0)

compare_and_swap()

```
boolean compare_and_swap( int *value,
                          int expected,
                          int new_value )
{
  int temp = *value;
  if ( *value == expected )
    *value = new_value;
  return temp;
}
```

Mutual-exclusion by compare_and_swap()

```
do {
  while( compare_and_swap( &lock, 0, 1 ) != 0 ) ;
     /* critical section code is here  */
  lock = 0;
     /* remainder section code is here */
} while (true};
```

Swaping the contents of two words (`compare_and_swap()`) on a shared variable (`lock`) (initially `lock=0`)

compare_and_swap()

```
boolean compare_and_swap( int *value,
                          int expected,
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{
  int temp = *value;
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  return temp;
}
```

Mutual-exclusion by compare_and_swap()

```
do {
  while( compare_and_swap( &lock, 0, 1 ) != 0 ) ;
     /* critical section code is here  */
  lock = 0;
     /* remainder section code is here */
} while (true};
```

The solution `compare_and_swap()` above is not bounded waiting.

Process P_i

```
while (true) {
  waiting[i] = true;
  key = 1;
  while (waiting[i] && key == 1)
    key = compare and swap(&lock,0,1);
  waiting[i] = false;

  /* critical section */

  j = (i + 1) % n;
  while ((j != i) && !waiting[j])
    j = (j + 1) % n;

  if (j == i)
    lock = 0;
  else
     waiting[j] = false;

  /* remainder section */
}
```

- `waiting` initialized to `false`, `lock` initialized to 0.
- On leaving CS, P_i scans in the cyclic ordering $(i+1, i+2, \ldots, n-1, 0, \ldots, i-1)$ (that `waiting[j] == true`), and then set `waiting[j] ==` `false` to allow P_j enter CS.
  $\Rightarrow$ Bounded waiting.

Atomic variables

Increment and decrement an integer value may produce race condition.

- `compare_and_swap()` is used as **basic** building blocks for constructing synchronization tools.
- **Atomic variable**: providing atomic operations (`increment()`, `decrement()`) on basic data types such as integers and booleans.

```
void increment(atomic int *v)
{
  int temp;
  do {
    temp = *v;
  } while (temp != compare and swap(v, temp, temp+1));
}
```

Outline

1. Basic concepts
2. Critical-section (CS) problem
3. Synchronization hardware
4. Software tools for synchronization
   - Mutex lock
   - Semaphores
5. Synchronization problems

Mutex locks

Application programmers do not like using hardware-synchronization mechanisms.

OS designers must provide tools to deal with critical-section with 2 functions

- `acquire()`

```
acquire{} {
  while (!available) ; /* busy wait */
  available = false;
}
```

- `release()`

```
release() {
  available = true;
}
```

Before entering CS, programmers call `acquire()`, and call `release()` when leaving CS. Both are atomic which can be implemented by hardware synchronization instructions.

Mutex locks

Application programmers do not like using hardware-synchronization mechanisms.

OS designers must provide tools to deal with critical-section with 2 functions

- `acquire()`

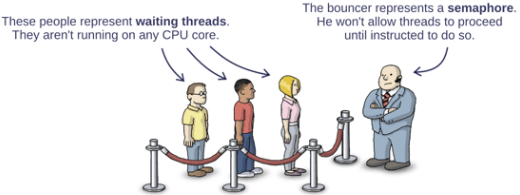
```
acquire{} {
  while (!available) ; /* busy wait */
  available = false;
}
```

- `release()`

Busy waiting = when a process is in CS, other processes loop continuously in order to enter CS (spinlock)

Before entering CS, programmers call `acquire()`, and call `release()` when leaving CS. Both are atomic which can be implemented by hardware synchronization instructions.

Semaphores

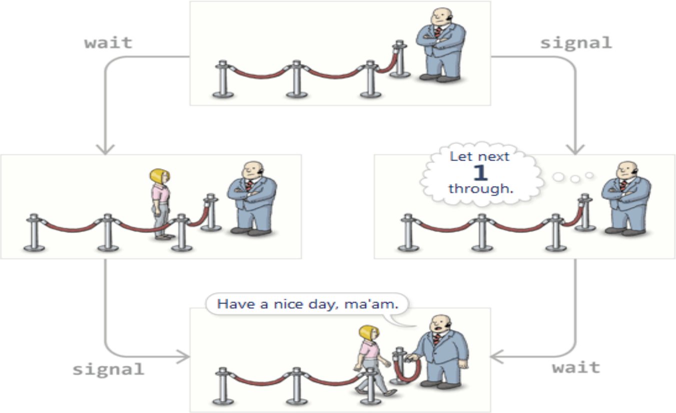


(source:preshing.com)

- Semaphore S: an integer variable
- can only be accessed by indivisible (atomic) operations

```
wait(S) {
  while ( S <= 0 ) ; /* busy wait */
  S --;
}
```

```
signal(S) {
  S ++;
}
```



(source:preshing.com)

Same outcome for different orders of calling `wait()` and `signal()`.

Semaphore usage

- Binary semaphore ($\{0,1\}$): similar to mutex locks
- Counting semaphore: to control access to resource with a finite number of instances
  - `S` initialized to number of resource instances
  - Wish to use an instance, call `wait(S)`
  - Release the resource, call `signal(S)`
  - All instances are in use, processes blocked until `S` > 0

Example of using semaphore `synch` to synchronize 2 processes (similar to `memory_barrier()`).

```
synch = 0;
```

P_1

```
S1;
signal( synch );
```

P_2

```
wait( synch );
S2;
```

Semaphores without busy waiting

Semaphore

Semaphore mentioned before could have busy waiting

Idea to avoid busy waiting

- After the checking of the semaphore fails, the process will be put into waiting queue
- The process is restarted by `wakeup()` when other processes send `signal()` (i.e., put the process into ready queue)

Semaphore data structure

```
typedef struct {
  int value;
  struct process *list;
} semaphore;
```

wait()

```
wait( semaphore *S ){
  S->value --;
  if ( S->value < 0 ){
    add this process to S->list;
    sleep();
  }
}
```

signal()

```
signal( semaphore *S ){
  S->value ++;
  if ( S->value >= 0 ) {
    remove a process P from S->list;
    wakeup( P );
  }
}
```

- block(), wakeup(): system calls

Semaphore data structure

```
typedef struct {
  int value;
  struct process *list;
} semaphore;
```

wait()

```
wait( semaphore *S }{
  S->value --;
}
```

signal()

```
signal( semaphore *S ){
  S->value ++;
```

Remarks

- Busy waiting just moved into critical section, not completely eliminated
- `wait()`, `signal()` must be executed atomically (on SMP, other techniques could be used additionally)

What is deadlock ?

P_0

```
wait(S);
wait(Q);
 ...
signal(S);
signal(Q);
```

P_1

```
wait(Q);
wait(S);
 ...
signal(Q);
signal(S);
```

- Two semaphores `S` and `Q` initialized to 1
- After `wait(S)` of P_0 and `wait(Q)` of P_1 executed, both processes wait for corresponding `signal()`s.

What is deadlock ?

P_0

```
wait(S);
wait(Q);
 ...
signal(S);
signal(Q);
```

P_1

```
wait(Q);
wait(S);
 ...
signal(Q);
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```

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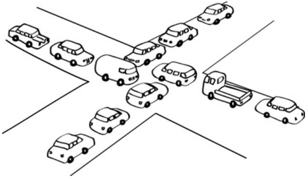

Deadlock

Two or more processes are waiting indefinitely for an event that can be caused by only one of the waiting processes.

livelock

What is deadlock ?

P_0

```
wait(S);
wait(Q);
 ...
signal(S);
signal(Q);
```

P_1

```
wait(Q);
wait(S);
 ...
signal(Q);
signal(S);
```

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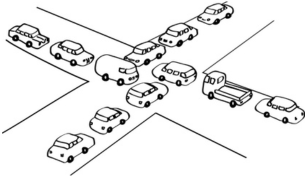

Deadlock

Two or more processes are waiting indefinitely for an event that can be caused by only one of the waiting processes.

Starvation (indefinite blocking)

A process may never be removed from the semaphore queue in which it is suspended

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Classic synchronization problems

- Only classic problems considered
  - Bounded-buffer problem
  - Readers-writers problem
  - Dining-philosophers problem
- Use semaphore for synchronization

Bounded-buffer problem

Semaphore data structure

```
int n;
semaphore mutex = 1;
semaphore empty = n;
semaphore full = 0;
```

producer

```
do {
  /* produce an item in      */
  /* next produced           */
  wait( empty );
  wait( mutex );
  /* add next_produced to    */
  /* the buffer              */
  signal( mutex );
  signal( full );
} while (true);
```

consumer

```
do {
  wait( full );
  wait( mutex );
  /* remove an item from the */
  /* buffer to next consumed */
  signal( mutex );
  signal( empty );
  /* consume the item in     */
  /* next_consumed           */
} while (true);
```

Readers-writers problem

- A database shared among several processes: some are readers, some are writers
- If a writer and some other process access database simultaneously chaos may occur

First readers-writers problem

no reader kept waiting unless a writer has already obtained permission to use shared objects

```
semaphore rw_mutex = 1;
semaphore mutex = 1;
int read_count = 0;
```

Writer

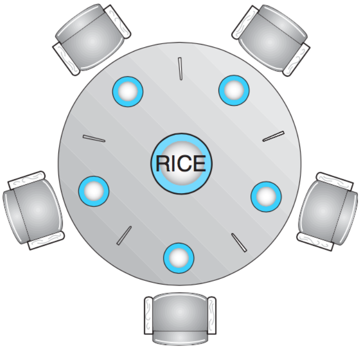
```
do {
  wait( rw_mutex );
  .../* writing performed */...
  signal( rw_mutex );
} while (true);
```

Reader

```
do {
  wait( mutex );
  read_count ++;
  if ( read_count == 1 )
    wait( rw_mutex );
  signal( mutex );
  .../* reading performed */...
  wait( mutex );
  read_count --;
  if ( read_count == 0 )
    signal( rw_mutex );
  signal( mutex );
} while (true);
```

Dining-philosopher

- Resource: 5 single chopsticks
- 5 Philosophers sitting around the table, with actions
  - Think: do nothing, not critical
  - Eat: need 2 chopsticks (left and right)



Semaphores can be used to solve dininig-philosophers synchronization problem, using 5 semaphores

```
semaphore chopstick[5];
```

Philosopher *i*

```
do {
  wait( chopstock[i] );
  wait( chopstick[(i+1) % 5] );
  /* eat eat eat */
  signal( chopstick[i] );
  signal( chopstick[(i+1) % 5] );
  /* think think think */
} while (true);
```